

API Descriptions

To support its various functionalities, **Roomies** would likely integrate with several APIs, including mapping, scheduling, and communication services.

1. Google Maps API

- **Functionality:** Helps users find nearby grocery stores, supermarkets, and essential services based on their location.
- **Use Case:** If a roommate adds an item to a grocery list, the app could suggest the closest store or an optimal route for completing errands.

2. Google Calendar API

- **Functionality:** Syncs with users' schedules to track availability for chores, meetings, or shared activities.
- **Use Case:** Automatically schedules cleaning tasks during times when all roommates are available.

3. Twilio API (Messaging & Notifications)

- **Functionality:** Enables SMS, in-app messaging, and push notifications.
- **Use Case:** Sends reminders about pending chores, due rent payments, or grocery pickups.

4. IBM Watson Personality Insights API

- **Functionality:** Analyzes user responses to personality quizzes and predicts compatibility between roommates.
- **Use Case:** Helps match roommates based on shared habits and communication styles.

5. OpenAI API (ChatGPT & AI Recommendations)

- **Functionality:** Generates AI-powered conflict resolution suggestions and personalized recommendations.
- **Use Case:** Detects when a roommate is frequently complaining about noise and offers diplomatic suggestions for discussion.

6. Plaid API (Financial Tracking & Payments)

- **Functionality:** Connects with users' bank accounts for easy bill splitting and shared expenses.
- **Use Case:** Roommates can track shared expenses, split rent, or handle grocery payments through the app.

By integrating these APIs, **Roomies** can enhance automation, improve roommate matching accuracy, and simplify household coordination, creating a seamless co-living experience.